$$V = \pi r^2 h + \left(\pi r^2 \times r - \frac{2}{3}\pi r^3\right) \Rightarrow V = \pi r^2 h + \frac{1}{3}\pi r^3$$

The upward force supports the weight of the liquid column above the free surface, therefore,

$$2\pi r T\cos\theta = \pi r^2\left(h + \frac{1}{3}r\right)\rho g \Rightarrow T = \frac{r\left(h + \frac{1}{3}r\right)\rho g}{2\cos\theta}$$

If the capillary is a very fine tube of radius (i.e., radius is very small) then $\frac{r}{3}$ can be neglected when it is compared to the height h. Therefore,

$$T = \frac{r\rho g h}{2\cos\theta} \qquad (7.33)$$

Liquid rises through a height h

$$h = \frac{2T\cos\theta}{r\rho g} \Rightarrow h \propto \frac{1}{r} \qquad (7.34)$$

This implies that the capillary rise (h) is inversely proportional to the radius (r) of the tube. i.e, the smaller the radius of the tube greater will be the capillarity.

EXAMPLE 7.12

Water rises in a capillary tube to a height of 2.0cm. How much will the water rise through another capillary tube whose radius is one-third of the first tube?

Solution

From equation (7.34), we have

$$h \propto \frac{1}{r} \Rightarrow hr = \text{constant}$$

Consider two capillary tubes with radius r_1 and r_2 which on placing in a liquid, capillary rises to height h_1 and h_2, respectively. Then,

$$h_1 r_1 = h_2 r_2 = \text{constant}$$

$$\Rightarrow\ h_2 = \frac{h_1 r_1}{r_2} = \frac{\left(2\times10^{-2}m\right)\times r}{\frac{r}{3}} \Rightarrow h_2 = 6\times10^{-2}\text{m}$$

EXAMPLE 7.13

Mercury has an angle of contact equal to 140° with soda lime glass. A narrow tube of radius 2 mm, made of this glass is dipped in a trough containing mercury. By what amount does the mercury dip down in the tube relative to the liquid surface outside? Surface tension of mercury T=0.456 N m^{-1}; Density of mercury ρ = 13.6 × 10^3 kg m^{-3}

Solution

Capillary descent, cos140 = cos(90+50) −sin50 = −0.7660

$$h = \frac{2T\cos\theta}{r\rho g} = \frac{2\times(0.456\,N\,m^{-1})(\cos140^0)}{(2\times10^{-3}m)(13.6\times10^3)(9.8\,ms^{-2})}$$

$$= \frac{2\times0.456\times(-0.7660)}{2\times13.6\times9.8}$$

$$= \frac{-0.6986}{266.56} = -2.62\times10^{-3}m$$

where, negative sign indicates that there is fall of mercury (mercury is depressed) in glass tube.

7.5.8 Applications of surface tension

- Mosquitoes lay their eggs on the surface of water. To reduce the surface tension of water, a small amount of oil is poured. This breaks the elastic film of water surface and eggs are killed by drowning.
- Chemical engineers must finely adjust the surface tension of the liquid, so it forms droplets of designed size and so it adheres to the surface without smearing. This is used in desktop printing, to paint automobiles and decorative items.
- Specks of dirt get removed when detergents are added to hot water while washing clothes because surface tension is reduced.
- A fabric can be made waterproof, by adding suitable waterproof material (wax) to the fabric. This increases the angle of contact.

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